so adds extra complexity, so we recommend going with one of the available runtimes.

Next, we will compare AWS Lambda with EC2 virtual machines.

### *6.1.3 Comparing AWS Lambda with virtual machines (Amazon EC2)*

What is the difference between using AWS Lambda and virtual machines? First is the granularity of virtualization. Virtual machines provide a full operating system for running one or multiple applications. In contrast, AWS Lambda offers an execution environment for a single function, a small part of an application.

Furthermore, Amazon EC2 offers virtual machines as a service, but you are responsible for operating them in a secure, scalable, and highly available way. Doing so requires you to put a substantial amount of effort into maintenance. In contrast, when building with Lambda, AWS manages the underlying infrastructure for you and provides a production-ready infrastructure.

Beyond that, AWS Lambda is billed per execution, and not per second like when a virtual machine is running. You don’t have to pay for unused resources that are waiting for requests or tasks. For example, running a script to check the health of a website every five minutes on a virtual machine would cost you about $3.71 per month. Executing the same health check with AWS Lambda will cost about $0.04 per month.

Table 6.1 compares AWS Lambda and virtual machines in detail. You’ll find a discussion of AWS Lambda’s limitations at the end of the